

ASSIGNMENT NO.9: Data Retrieval, Functions, and Joins in MySQL

Q1) Create a MySQL database named CollegeDB and create two tables: OnlineEnrollment and OfflineEnrollment.

OnlineEnrollment Table

Column Name	Data Type	Constraint
StudentID	INT	PRIMARY KEY
StudentName	VARCHAR(50)	NOT NULL
Course	VARCHAR(30)	
City	VARCHAR(30)	
Marks	INT	

OfflineEnrollment Table

Column Name	Data Type	Constraint
StudentID	INT	PRIMARY KEY
StudentName	VARCHAR(50)	NOT NULL
Course	VARCHAR(30)	

USE CollegeDB;

```
CREATE TABLE OnlineEnrollment (  
    StudentID INT PRIMARY KEY,  
    StudentName VARCHAR(50) NOT NULL,  
    Course VARCHAR(30),  
    City VARCHAR(30),  
    Marks INT  
);
```

```
CREATE TABLE OfflineEnrollment (  
    StudentID INT PRIMARY KEY,  
    StudentName VARCHAR(50) NOT NULL,  
    Course VARCHAR(30),  
    City VARCHAR(30),
```

Marks INT

);

INSERT INTO OnlineEnrollment

(StudentID, StudentName, Course, City, Marks)

VALUES

(1, 'Rahul Sharma', 'BSc IT', 'Mumbai', 78),

(2, 'Priya Patil', 'BCA', 'Pune', 85),

(3, 'Aman Verma', 'BSc CS', 'Nashik', 67),

(4, 'Sneha Joshi', 'BSc IT', 'Mumbai', 91),

(5, 'Karan Mehta', 'BCA', 'Thane', 74),

(6, 'Neha Shah', 'BSc IT', 'Pune', 88);

INSERT INTO OfflineEnrollment

(StudentID, StudentName, Course, City, Marks)

VALUES

(4, 'Sneha Joshi', 'BSc IT', 'Mumbai', 91),

(5, 'Karan Mehta', 'BCA', 'Thane', 74),

(7, 'Rohan Desai', 'BSc CS', 'Mumbai', 69),

(8, 'Meera Kulkarni', 'BCA', 'Nashik', 82),

(9, 'Arjun Nair', 'BSc IT', 'Thane', 59),

(10, 'Divya More', 'BCA', 'Mumbai', 93);

phpMyAdmin

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Server: 127.0.0.1 » Database: collegedb » Table: OfflineEnrollment

Browse Structure SQL Search Insert Export Import Privile

Show query box

Showing rows 0 - 5 (6 total, Query took 0.0008 seconds.)

SELECT * FROM OfflineEnrollment;

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	StudentID	StudentName	Course	City	Marks
☐ Edit Copy Delete	1	Rahul Sharma	BSc IT	Mumbai	78
☐ Edit Copy Delete	2	Priya Patil	BCA	Pune	85
☐ Edit Copy Delete	3	Aman Verma	BSc CS	Nashik	67
☐ Edit Copy Delete	4	Sneha Joshi	BSc IT	Mumbai	91
☐ Edit Copy Delete	5	Karan Mehta	BCA	Thane	74
☐ Edit Copy Delete	6	Neha Shah	BSc IT	Pune	88

Check all With selected: Edit Copy Delete Export

Console

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Server: 127.0.0.1 » Database: collegedb » Table: OfflineEnrollment

Browse Structure SQL Search Insert Export Import Privileges Open

Showing rows 0 - 5 (6 total, Query took 0.0005 seconds.)

SELECT * FROM OfflineEnrollment;

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	StudentID	StudentName	Course	City	Marks
☐ Edit Copy Delete	4	Sneha Joshi	BSc IT	Mumbai	91
☐ Edit Copy Delete	5	Karan Mehta	BCA	Thane	74
☐ Edit Copy Delete	7	Rohan Desai	BSc CS	Mumbai	69
☐ Edit Copy Delete	8	Meera Kulkarni	BCA	Nashik	82
☐ Edit Copy Delete	9	Arjun Nair	BSc IT	Thane	59
☐ Edit Copy Delete	10	Divya More	BCA	Mumbai	93

Check all With selected: Edit Copy Delete Export

Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

2) Write a MySQL query using UNION to display all unique students from both OnlineEnrollment and OfflineEnrollment.

The output should contain:

Column
StudentID
StudentName
Course
City
Marks

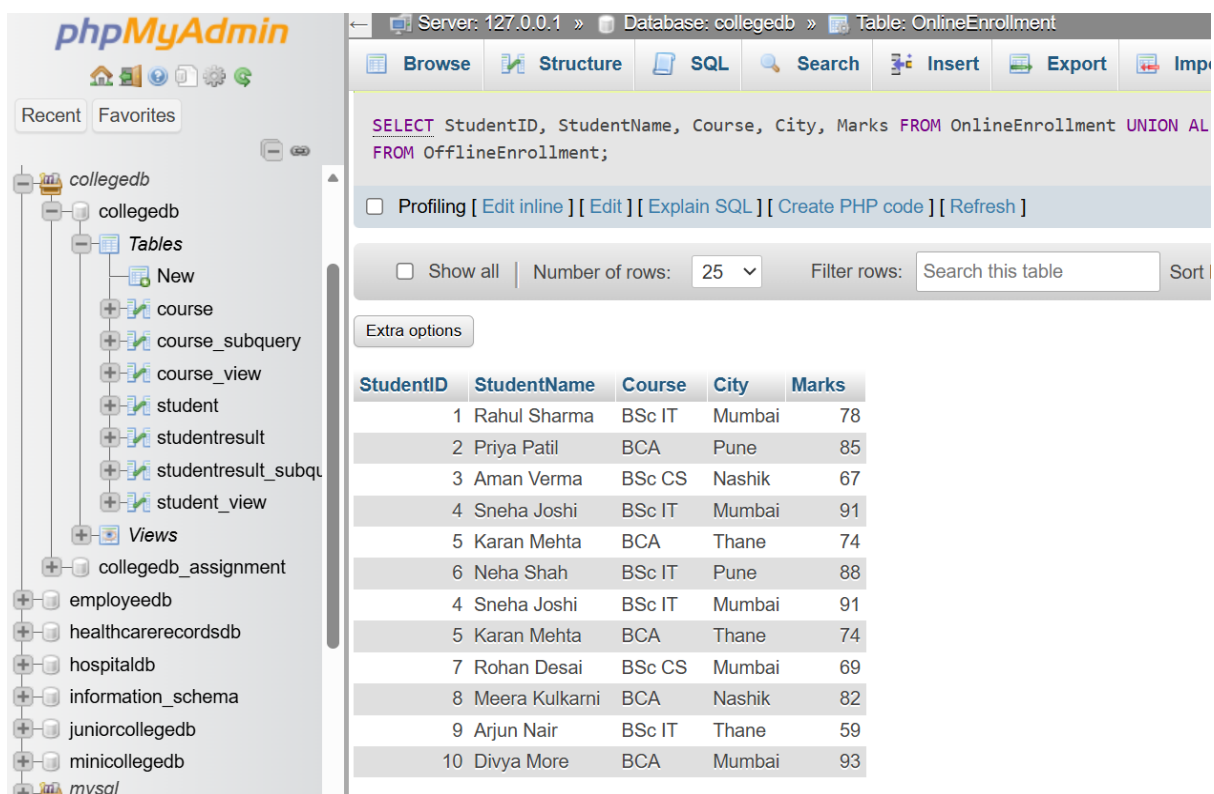
```
SELECT StudentID, StudentName, Course, City, Marks
FROM OnlineEnrollment
UNION
SELECT StudentID, StudentName, Course, City, Marks
FROM OfflineEnrollment;
```

The screenshot shows the phpMyAdmin interface. On the left, the database structure for 'collegedb' is visible, including tables like 'course', 'student', and 'studentresult'. The main panel shows the SQL query: `SELECT StudentID, StudentName, Course, City, Marks FROM OnlineEnrollment UNION OfflineEnrollment;`. The results are displayed in a table with the following data:

StudentID	StudentName	Course	City	Marks
1	Rahul Sharma	BSc IT	Mumbai	78
2	Priya Patil	BCA	Pune	85
3	Aman Verma	BSc CS	Nashik	67
4	Sneha Joshi	BSc IT	Mumbai	91
5	Karan Mehta	BCA	Thane	74
6	Neha Shah	BSc IT	Pune	88
7	Rohan Desai	BSc CS	Mumbai	69
8	Meera Kulkarni	BCA	Nashik	82
9	Arjun Nair	BSc IT	Thane	59
10	Divya More	BCA	Mumbai	93

Q3) Write a MySQL query using UNION ALL to combine records from both tables while keeping duplicate records.

```
SELECT StudentID, StudentName, Course, City, Marks
FROM OnlineEnrollment
UNION ALL
SELECT StudentID, StudentName, Course, City, Marks
FROM OfflineEnrollment;
```



The screenshot shows the phpMyAdmin interface for a database named 'collegedb'. The SQL editor displays the following query:

```
SELECT StudentID, StudentName, Course, City, Marks FROM OnlineEnrollment UNION ALL
FROM OfflineEnrollment;
```

Below the query editor, the 'Extra options' section is expanded, showing a table with 10 rows of data. The table has the following columns: StudentID, StudentName, Course, City, and Marks.

StudentID	StudentName	Course	City	Marks
1	Rahul Sharma	BSc IT	Mumbai	78
2	Priya Patil	BCA	Pune	85
3	Aman Verma	BSc CS	Nashik	67
4	Sneha Joshi	BSc IT	Mumbai	91
5	Karan Mehta	BCA	Thane	74
6	Neha Shah	BSc IT	Pune	88
4	Sneha Joshi	BSc IT	Mumbai	91
5	Karan Mehta	BCA	Thane	74
7	Rohan Desai	BSc CS	Mumbai	69
8	Meera Kulkarni	BCA	Nashik	82
9	Arjun Nair	BSc IT	Thane	59
10	Divya More	BCA	Mumbai	93

Q4) Write a MySQL query using INTERSECT to display students who are present in both online and offline enrollment tables.

The output should contain:

Column
StudentID
StudentName
Course
City
Marks

```
SELECT StudentID, StudentName, Course, City, Marks  
FROM OnlineEnrollment
```

INTERSECT

```
SELECT StudentID, StudentName, Course, City, Marks  
FROM OfflineEnrollment;
```

The screenshot shows the phpMyAdmin interface. On the left is a sidebar with a database tree. The main panel displays a SQL query and its results. The query is an SQL query using the INTERSECT operator. The results table shows two rows of data.

Database: collegedb » Table: OnlineEnrollment

Showing rows 0 - 1 (2 total, Query took 0.0011 seconds.)

```
SELECT StudentID, StudentName, Course, City, Marks FROM OnlineEnrollment INTERSECT SE  
FROM OfflineEnrollment;
```

StudentID	StudentName	Course	City	Marks
4	Sneha Joshi	BSc IT	Mumbai	91
5	Karan Mehta	BCA	Thane	74

Q5) Write a MySQL query using EXCEPT to display students who are present only in the online enrollment table.

The output should contain:

Column
StudentID
StudentName
Course
City
Marks

```
SELECT StudentID, StudentName, Course, City, Marks  
FROM OnlineEnrollment
```

EXCEPT

```
SELECT StudentID, StudentName, Course, City, Marks  
FROM OfflineEnrollment;
```

The screenshot shows the phpMyAdmin interface. On the left is a sidebar with a database tree. The main panel displays a SQL query in the 'SQL' tab:

```
SELECT StudentID, StudentName, Course, City, Marks FROM OnlineEnrollment EXCEPT SELECT StudentID, StudentName, Course, City, Marks FROM OfflineEnrollment;
```

Below the query editor, there are controls for 'Show all', 'Number of rows' (set to 25), 'Filter rows' (Search this table), and 'Sort by key' (None). The 'Extra options' section is collapsed. The 'Query results operations' section is expanded, showing buttons for 'Print', 'Copy to clipboard', 'Export', 'Display chart', and 'Create view'.

StudentID	StudentName	Course	City	Marks
1	Rahul Sharma	BSc IT	Mumbai	78
2	Priya Patil	BCA	Pune	85
3	Aman Verma	BSc CS	Nashik	67
6	Neha Shah	BSc IT	Pune	88

Q6) Write a MySQL query to display high-scoring students from both tables.

```
SELECT StudentID, StudentName, Course, Marks  
FROM OnlineEnrollment  
WHERE Marks >= 85  
UNION  
SELECT StudentID, StudentName, Course, Marks  
FROM OfflineEnrollment  
WHERE Marks >= 85  
ORDER BY Marks DESC;
```

phpMyAdmin

Server: 127.0.0.1 » Database: collegedb » Table: OnlineEnrollment

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studentresult

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Views

collegedb_assignment

employeedb

healthcarerecordsdb

hospitaldb

information_schema

juniorcollegedb

minicollegedb

mysql

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Showing 1 row(s) of 1 row(s) query result (0.000000 seconds)

```
SELECT StudentID, StudentName, Course, Marks FROM OnlineEnrollment WHERE Marks >= 85 UNION SELECT StudentID, StudentName, Course, Marks FROM OfflineEnrollment WHERE Marks >= 85 ORDER BY Marks DESC;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

StudentID	StudentName	Course	Marks
10	Divya More	BCA	93
4	Sneha Joshi	BSc IT	91
6	Neha Shah	BSc IT	88
2	Priya Patil	BCA	85

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

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